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# Improving Antimicrobial Resistance (AMR) Phenotype Prediction for Unseen Bacteria through Data Augmentation and Machine Learning

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## Abstract

Antimicrobial resistance (AMR) poses a growing threat to global health, yet accurately predicting resistance remains challenging due to limited and imbalanced genomic data. Existing machine learning (ML) models often struggle to generalize across diverse genome–antibiotic combinations, especially in data-scarce environments. AMR prediction is hindered by the high cost and time required for experimental data collection, as well as the lack of sufficient labeled genomic datasets. Additionally, current ML models face limitations in performance and generalizability when applied to small or imbalanced datasets. To address these challenges, we propose  $k$ -mer-based feature extraction for ML models. We further enhance model performance and generalizability through data augmentation techniques, enabling accurate AMR prediction despite limited training data.

Our experimental results show that tree-based models with hyperparameter tuning perform well on small datasets, while CNN models trained with augmented data achieve superior accuracy and robustness across diverse genome–antibiotic pairs. Notably, the proposed deep 1D CNN framework achieves performance that equals or surpasses prior benchmarks, underscoring the benefit of coupling data augmentation with deep learning for AMR prediction. Furthermore, we validate that the proposed approach can successfully generalize to newly identified bacteria by accurately classifying multiple antibiotic classes in a multiclass prediction setting. These findings demonstrate that tree-based ensemble models enable rapid and reliable predictions on small datasets, while deep learning models achieve higher performance by leveraging data augmentation to address data scarcity. This study establishes a foundation for reducing antibiotic misuse by enabling accurate antibiotic selection, even for newly identified and previously unseen bacterial strains.

## CCS Concepts

• **Computing methodologies** → **Supervised learning**; **Neural networks**; • **Computational biology** → **Genomics**; **Molecular sequence analysis**; • **Applied computing** → **Health informatics**.

\*Both authors contributed equally to this research.



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## Keywords

Antimicrobial Resistance (AMR), Machine Learning, Data Augmentation, Convolutional Neural Network (CNN),  $K$ -mer Features

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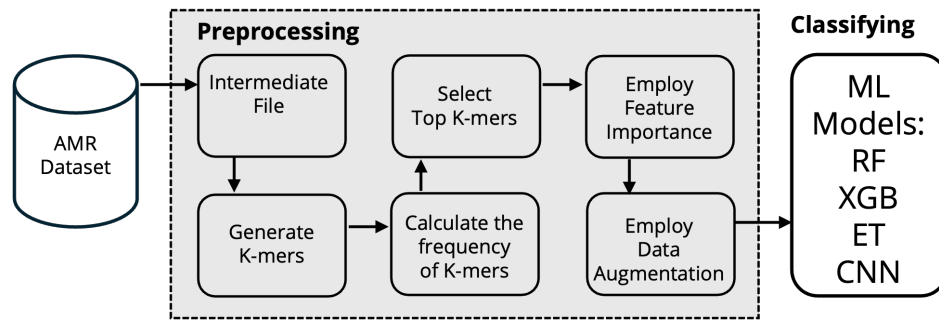
## 1 Introduction

This study aims to improve the predictive performance of antimicrobial resistance (AMR) classification models. This is achieved by integrating advanced machine learning (ML) algorithms with data augmentation techniques to address the challenges of small and imbalanced genomic datasets. AMR is an urgent and growing global health threat, contributing to a rise in hospital-acquired infections and increased mortality, especially among immunocompromised and oncology patients. If not addressed, AMR is projected to surpass cancer as a leading cause of death by 2050 [5, 24]. Many researchers have built databases such as BV-BRC<sup>1</sup> and are actively conducting research using this data, leveraging advances in machine learning (ML) [17, 26].

Despite advancements in ML for AMR prediction, several critical challenges continue to hinder progress. One major issue is that extracting and preprocessing AMR datasets from genomic and clinical sources remains labor intensive and time consuming [6, 10]. Second, many genome–antibiotic combinations suffer from a lack of sufficient labeled data, which impedes the development of generalizable predictive models [2]. The availability and quality of data are major challenges in AMR research, especially for less common microbes or those isolated from unusual environments [13]. This problem is more pronounced in small laboratories with limited data, where models are more prone to overfitting due to the low sample size. Finally, despite advances in ML for AMR prediction, performance remains limited by data imbalance, sparse features, and poor generalizability. Models trained on imbalanced data may favor the majority class, and low-quality data can further reduce accuracy [11]. Addressing these challenges is essential to building more robust and clinically relevant AMR prediction systems.

To address the aforementioned challenges in AMR prediction, this study introduces a multi-faceted approach that combines fine-grained classification pipelines with deep learning models enhanced

<sup>1</sup><https://www.bv-brc.org/>



**Figure 1: AMR Classification Framework Using  $K$ -mer Features and Machine Learning**

by data augmentation. First, to reduce the complexity of raw genomic data and extract meaningful patterns, we adopt a feature extraction method based on  $k$ -mers (also known as  $n$ -gram), which involves tokenizing genomic sequences into fixed-length substrings [9]. Second, to mitigate the issue of limited and imbalanced data, we integrate data augmentation techniques [7]. This helps synthetically balance class distributions, improving the representation of minority classes and preventing model bias toward the majority class. In addition, in cases where deep learning models—particularly Convolutional Neural Network (CNN) models—underperform due to a lack of data, data augmentation can be used to enhance the effectiveness of the CNN model.

Figure 1 illustrates the preprocessing pipeline designed to transform genomic data into a format suitable for machine learning models in AMR prediction, which is discussed in detail in Section 3. The main contributions are described below:

- We analyze the effectiveness of tree-based models with hyperparameter optimization (via Bayesian, Randomized Search, and Grid Search), demonstrating that they can achieve accurate predictions even with low data availability, making them suitable for rapid AMR prediction in data-constrained settings (Section 4.1).
- This study addresses the limitations of small and imbalanced datasets by combining deep learning models (CNNs) with data augmentation techniques. This enables effective AMR prediction even when labeled data is scarce, which is a common challenge in genomic research (Section 4.2).
- We further design a multiclass prediction setting to evaluate the model’s ability to generalize to newly identified bacteria. It demonstrates that the proposed feature selection and modeling strategy can accurately classify multiple antibiotic classes even for unseen bacterial profiles (Section 4.3).

The structure of this paper is as follows. Section 2 provides background information highlighting the importance of AMR prediction and reviews existing research. Section 3 discusses the characteristics of the AMR dataset used in this study, and outlines the experimental procedures, including the algorithm used to generate  $k$ -mer-based features. Section 4 presents three experiments evaluating machine learning models, including tree-based ensemble methods with hyperparameter optimization, CNN models incorporating data augmentation, and a multiclass prediction setting designed to assess generalization to newly identified bacteria in

AMR dataset prediction. Section 5 discusses the strengths and limitations of tree-based ensembles, CNNs, and data augmentation in AMR prediction, focusing on trade-offs for building effective research and clinical pipelines. Lastly, Section 6 summarizes the conclusions of this study and discusses potential directions for future research.

## 2 Related Work

Antibiotic resistance is a major global threat, driving efforts to develop predictive models using machine learning [18]. These efforts improve AMR prediction by applying diverse algorithms including machine learning to complex biological data [16]. Nsubuga *et al.* [16] developed and validated machine learning models trained on whole-genome sequencing data from England to predict AMR. Wang *et al.* [27] used eICU Research Institute (eRI) database using ML models, such as Logistic Regression, Random Forest, and Gradient Boosting.

There are numerous studies that have utilized data from BV-BRC<sup>2</sup> [19, 21]. Numerous research studies have employed  $k$ -mer features as a foundational approach for representing genomic or sequence-based data [15, 22, 23]. Several research studies have utilized Single Nucleotide Polymorphism (SNP) features to represent genetic variation across individuals or populations [3, 28]. A wide range of machine learning algorithms have been employed in recent studies to analyze genomic and antimicrobial resistance (AMR) data. For instance, LASSO logistic regression selects key genomic features for AMR prediction by penalizing large coefficients [14]. Similarly, SVMs find optimal hyperplanes to separate resistant and susceptible samples in high-dimensional spaces [28]. Additionally, tree-based ensemble models, including Random Forests, Gradient Boosting Machines, and related methods, have gained widespread popularity for their ability to capture non-linear interactions and feature importance in complex genomic datasets [3, 12]. These ensemble approaches are particularly valued for their interpretability, robustness to overfitting, and high predictive accuracy in biological classification tasks. James *et al.* [26] conducted an extensive study on AMR prediction using data from the BV-BRC database. They utilized machine learning models to classify susceptible and resistant phenotypes based on AMR metadata collected from BV-BRC, an information system specifically designed to support research

<sup>2</sup><https://www.bv-brc.org/>

**Table 1: Dataset statistics for resistant and susceptible genome–antibiotic combinations.**

| Genome              | Antibiotic                          | Resistant |          |          | Susceptible |          |          |
|---------------------|-------------------------------------|-----------|----------|----------|-------------|----------|----------|
|                     |                                     | Count     | Avg (MB) | STD (MB) | Count       | Avg (MB) | STD (MB) |
| <i>E. faecium</i>   | Vancomycin (VAN)                    | 1,229     | 2.81     | 0.99     | 1,149       | 2.76     | 0.84     |
| <i>E. coli</i>      | Ampicillin (AMP)                    | 3,026     | 4.98     | 0.37     | 2,577       | 4.90     | 0.25     |
| <i>A. baumannii</i> | Amikacin (AMK)                      | 727       | 3.83     | 0.68     | 370         | 3.93     | 0.32     |
| <i>E. coli</i>      | Gentamicin (GEN)                    | 797       | 4.78     | 1.01     | 6,132       | 4.90     | 0.46     |
| <i>E. coli</i>      | Trimethoprim/Sulfamethoxazole (SXT) | 1,081     | 4.98     | 0.30     | 2,477       | 4.93     | 0.23     |
| <i>E. faecium</i>   | Ampicillin (AMP)                    | 531       | 2.97     | 0.54     | 209         | 2.81     | 0.60     |
| <i>S. enterica</i>  | Ampicillin (AMP)                    | 1,954     | 4.76     | 0.35     | 2,709       | 4.68     | 0.30     |
| <i>S. enterica</i>  | Chloramphenicol (CHL)               | 1,083     | 4.78     | 0.38     | 3,486       | 4.64     | 0.30     |

on bacterial and viral infectious diseases. Their approach used 7-mer nucleotide features, F1 scores for evaluation, and the XGBoost algorithm, leveraging over 75 resistant and susceptible genomes.

Compared to previous studies, our research uniquely combines fine-grained modeling with both tree-based and deep learning approaches tailored to specific genome–antibiotic pairs. Unlike prior work that often relied solely on large, curated datasets, we address data scarcity through effective data augmentation techniques. This allows our models, particularly the CNN-based framework, to achieve high performance even in imbalanced or limited-data scenarios, enhancing practical applicability.

### 3 Research Approach

This study utilizes a dataset obtained from the BV-BRC repository, which contains antimicrobial resistance (AMR) phenotype data derived from antimicrobial susceptibility testing (AST) reported in published literature and collaborative research initiatives. In the PATRIC\_genome\_AMR.txt<sup>3</sup> file, the “resistant\_phenotype” column includes a total of 167,916 entries labeled as resistant and 334,785 labeled as susceptible, with individual file sizes ranging from 2MB to 7MB. For this study, we have selected 8 different types of genome-antibiotic pairs to investigate by using the .fna files, as they contain only raw nucleic acid sequences. The goal is to perform binary classification, distinguishing between “resistant” and “susceptible” cases. Table 1 provides detailed statistics on the dataset used for AMR prediction. It outlines the genome-antibiotic pairs, along with sample counts, average genome sizes, and standard deviations (STD) for both resistant and susceptible strains. The “Genome Name” refers to the bacterial species analyzed in the study. The “Antibiotic” indicates the antimicrobial agent used to determine whether a bacterial strain is resistant or susceptible. The “Count / Avg / STD” values for both “Resistant and Susceptible” represent the number of genomes, their average size in megabytes (MB), and the standard deviation. Generally, the susceptible class has more samples than the resistant class. Genome sizes tend to be similar across most pairs. Variability is relatively low, suggesting consistent genome sizes across strains. This table confirms that the study covers multiple bacteria-antibiotic pairs with a range of genome sizes and resistance profiles. The imbalance between classes reinforces the need for data augmentation techniques.

This study employs an  $k$ -mer frequency-based analysis approach. Prior research has shown that  $k$ -mers of length 7 to 10 nucleotides are typically sufficient to achieve favorable trade-offs between computational efficiency and classification accuracy [26]. Building on previously published methods [4], we adapt our existing  $k$ -mer generation tools for use in this genomic context. This reuse of established infrastructure not only streamlined the implementation but also reinforced continuity with our earlier work. The following are the detailed steps involved in the experiment:

- Step 1. Download AMR dataset files from the BV-BRC repository to construct the sets  $R$  and  $S$ .
- Step 2. Select equal-sized subsets  $R' \subseteq R$  and  $S' \subseteq S$  such that  $|R'| = |S'| = N$ , where  $N$  is the desired number of samples per class.
- Step 3. For each file  $f \in R' \cup S'$ , filter out metadata to retain only the nucleotide sequences, denoted  $\text{seq}(f)$ .
- Step 4. From each sequence  $\text{seq}(f)$ , generate all contiguous substrings of length  $k = 10$  (i.e., 10-mers):

$$\mathcal{K}_f = \{\text{seq}(f)[i : i + 10] \mid 0 \leq i \leq |\text{seq}(f)| - 10\}$$

- Step 5. Aggregate  $k$ -mer frequencies across all files and compute the frequency function  $\text{freq}(k_i)$  for each unique  $k$ -mer  $k_i$ . Select the top  $F$  most frequent  $k$ -mers, where  $F \ll 10^4$  (e.g., 10,000).
- Step 6. Apply feature importance scores from a tree-based ensemble model (e.g., Random Forest) to refine the feature set:

$$F_{\text{selected}} = \arg\max_{F' \subseteq F_{\text{top}}} \text{FeatureImportance}(F')$$

where  $F_{\text{top}}$  denotes the top  $F$  most frequent  $k$ -mers selected in Step 5, and  $F_{\text{selected}}$  denotes the final set of selected features (e.g., 1,000 features) based on feature importance ranking.

- Step 7. Construct the feature matrix  $X \in \mathbb{R}^{2N \times |F_{\text{selected}}|}$ , where each row corresponds to a file and each column to the frequency of a selected  $k$ -mer. Define the label vector  $y \in \{0, 1\}^{2N}$  as:

$$y_i = \begin{cases} 0 & \text{if sample } i \in R' \\ 1 & \text{if sample } i \in S' \end{cases}$$

- Step 8. Apply data augmentation techniques (e.g., SMOTE, oversampling) to generate synthetic data that enhances class balance and model robustness. The validation (test) dataset

<sup>3</sup>[https://www.bv-brc.org/docs/quick\\_references/ftp.html](https://www.bv-brc.org/docs/quick_references/ftp.html)

remains untouched to prevent data leakage and ensure a fair evaluation.

- Step 9. Train ensemble machine learning models—such as Random Forest (RF), Extra Trees (XT), XGBoost (XGB), and CNN—on the training fold combined with the generated synthetic data, and evaluate their performance on the corresponding test fold containing only real (non-augmented) data.



**Figure 2: Transition from Step 3 to Step 4: Genomic Sequence (fna) to  $k$ -mer Conversion**

Figure 2 illustrates the conversion of genomic sequence data (fna file) into  $k$ -mers, marking the transition from Step 3 to Step 4 of the workflow. The left panel shows the original DNA sequence of *Enterococcus faecium* (genome\_id: 1352.10001), identified as a vancomycin-resistant strain, along with header metadata (e.g., “>1352...”). Prior to  $k$ -mer generation, this metadata is removed, and only the nucleotide sequence is retained. The right panel shows the resulting  $k$ -mers, obtained by sliding a fixed-length window ( $k$ ) across the sequence. This transformation simplifies raw genomic data and serves as a critical preprocessing step for downstream applications such as machine learning, genome comparison, and antimicrobial resistance prediction.

Model performance was evaluated using 5-fold cross-validation on the training folds, and the evaluation metrics (Accuracy, Precision, Recall, and F1-score) were calculated exclusively on separate test folds to prevent data leakage. In addition, paired  $t$ -tests were conducted to confirm the statistical significance of the performance improvements achieved through data augmentation. All experiments were conducted on a system equipped with an Apple M3 Pro processor (11 cores, 18 GB RAM) running macOS Sequoia 15.4.1. Deep learning models were implemented in PyTorch, and ensemble models in scikit-learn.

## 4 Experiment

### 4.1 Experiment 1 (EX1): Tree-Based Ensemble Models

Figure 3 illustrates the relationship between scaled frequencies and feature importance scores. The X-axis represents the top 10,000 features ranked by frequency. The Y-axis displays the feature importance index alongside the frequency, which has been scaled using a soft log function. In the figure, the red line represents frequency, while the black bars indicate the feature importance score for each feature. The red line illustrates a monotonically decreasing trend

in feature frequency, indicating that the most frequent features are concentrated at the beginning of the index range. The long tail of low-frequency features supports power-law distribution often seen in natural datasets like  $k$ -mer frequencies [25]. In contrast, the majority of features have very low importance, close to zero. A small number of spikes indicate features with significantly higher importance. This pattern suggests that only a few features are highly influential in the model’s performance, which is common in high-dimensional feature spaces like those in genomic or text data [1]. The correlation coefficient (CORREL) between frequency and feature importance is approximately 0.003, indicating a negligible linear relationship between these two variables. In other words, the feature importance values do not show a meaningful linear relationship with their frequency in this dataset. The black vertical spikes highlight that some low-frequency features still exhibit high importance. The top three features are: ttttataaaa (index 5584, score 0.0195), aaaacggca (index 9020, score 0.0102), and ccggtaccgg (index 8939, score 0.0063). These features are considered the most impactful for the model’s predictions. This suggests that high-frequency features do not necessarily have high importance, while certain low-frequency features can still play a significant role in prediction.

In EX1, we evaluate the performance of the algorithms described in Section 3, focusing on their effectiveness across various dataset sizes and feature dimensions. This experiment aims to provide a comprehensive understanding of how the proposed methods perform under realistic and potentially challenging preprocessing conditions. Prior studies have shown that increasing feature dimensionality or using a large volume of data during preprocessing can significantly increase computation time, which may become a bottleneck for practical deployment in real world settings [4]. To mitigate this issue, the study employed a fixed subset of 200 data points from each category and, as shown in Figure 3, selected 1,000 features from the top 10,000 most frequent features based on feature importance methods, aiming to reduce computational overhead without compromising model quality. Additionally, several parameter tuning techniques were incorporated to further enhance the predictive performance of the models. In particular, the tuning efforts were directed at optimizing the hyperparameters of tree-based ensemble models, which is known to be highly sensitive to configuration choices.

Table 2 presents model performance across hyperparameter tuning methods—Bayesian (BayesSearchCV), Randomized (Randomize dSearchCV), and Grid (GridSearchCV)—for various genome–antibiotic combinations, using three different machine learning models: Random Forest (RF), Extreme Gradient Boosting (XGB), and Extra Trees (ET).

All evaluations are performed on the “Original” dataset, which includes 200 resistant and 200 susceptible samples. The table reports only the best-performing machine learning model and hyperparameter tuning method for each genome–antibiotic pair. To provide a comparative benchmark, the final column reports performance metrics based on the work of VanOeffelen *et al.* [26]. This reference serves as a baseline for evaluating the effectiveness of the tuning strategies and models used in our study. Since their results were presented using color gradients rather than exact numerical values, we use the tilde symbol “~” to indicate that the reported values



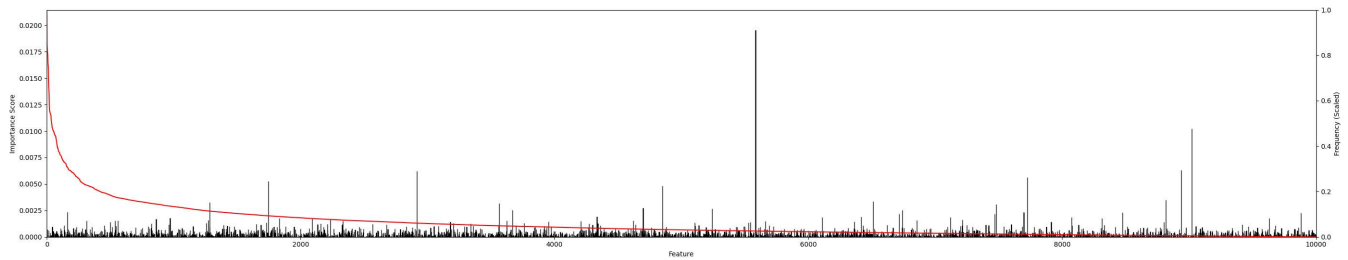


Figure 3: Frequency vs. Feature Importance for Salmonella - CHL

Table 2: Model performance across different hyperparameter tuning methods compared with prior work by VanOeffelen *et al.* [26].

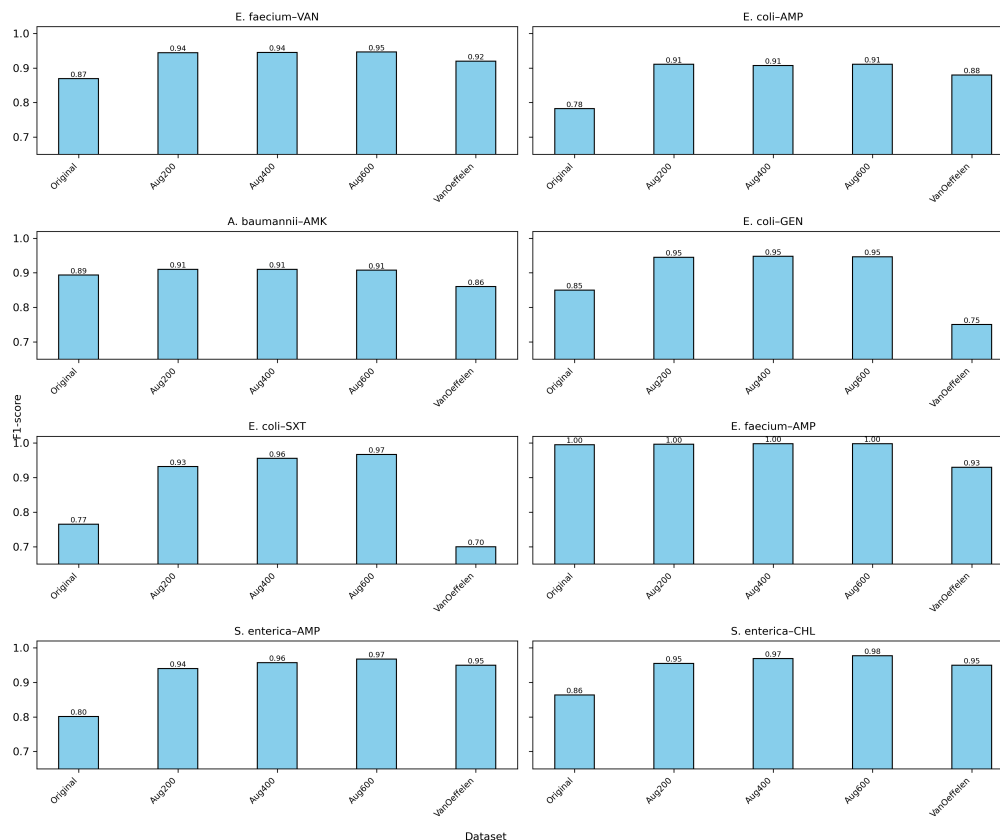
| Genome              | Antibiotic                          | Model | Tuning Method | F1-score | VanOeffelen <i>et al.</i> |
|---------------------|-------------------------------------|-------|---------------|----------|---------------------------|
| <i>E. faecium</i>   | Vancomycin (VAN)                    | XGB   | Bayesian      | 0.9231   | ~92% (XGB)                |
| <i>E. coli</i>      | Ampicillin (AMP)                    | XGB   | Randomized    | 0.8963   | ~88% (XGB)                |
| <i>A. baumannii</i> | Amikacin (AMK)                      | ET    | Grid          | 0.9042   | ~86% (XGB)                |
| <i>E. coli</i>      | Gentamicin (GEN)                    | ET    | Randomized    | 0.9828   | ~75% (XGB)                |
| <i>E. coli</i>      | Trimethoprim/Sulfamethoxazole (SXT) | ET    | Bayesian      | 0.7500   | ~70% (XGB)                |
| <i>E. faecium</i>   | Ampicillin (AMP)                    | RF    | Randomized    | 1.0000   | ~93% (XGB)                |
| <i>S. enterica</i>  | Ampicillin (AMP)                    | RF    | Grid          | 0.8776   | ~95% (XGB)                |
| <i>S. enterica</i>  | Chloramphenicol (CHL)               | XGB   | Grid          | 0.8718   | ~95% (XGB)                |

are approximate, based on our visual estimation. In addition, their work used nucleotide 7-mers as input features [26]. This work generally improves or matches prior benchmarks, particularly when using Randomized or Grid Search with XGBoost. In six out of eight genome-antibiotic combinations, our pipeline either outperforms or closely matches the F1-score reported by VanOeffelen *et al.* [26]. This is particularly evident in the case of *E. coli* treated with GEN, where our model achieved an F1-score of 98.28%, representing a substantial improvement over the ~75% reported in their study. Similarly, for *A. baumannii* treated with AMK, our model reached an F1-score of 90.42%, which is notably higher than their reported performance of approximately 86%. This table indicates the best-performing combinations of machine learning models and hyperparameter tuning methods across various genome-antibiotic pairs. No single hyperparameter tuning method dominates across all tasks. Grid, Randomized, and Bayesian methods each appear in top-performing configurations, emphasizing the importance of customizing tuning strategies for each task. Overall, EX1 underscores the importance of model-tuning customization rather than a one-size-fits-all approach, and it validates the robustness of the current study’s methods against established benchmarks. has shown that increasing dataset size does not always translate to substantial performance gains for tree-based ensemble models. Building on this finding, the current experiment focuses on evaluating the performance of deep learning models, which are generally more suitable for handling larger datasets.

## 4.2 Experiment 2 (EX2): CNN Model

To enhance predictive performance, EX2 leverages a larger dataset in conjunction with deep learning techniques, specifically Convolutional Neural Networks (CNNs). CNNs are particularly well-suited for learning complex patterns from high-dimensional biological data due to their ability to automatically extract hierarchical features through convolutional layers [20]. Instead of adding more dataset for training which may increase pre-process overhead, this study incorporates data augmentation strategies [8]. EX2 increases the variability and representativeness of the training data, enabling the CNN model to generalize better across different genome-antibiotic combinations. This approach not only improves classification accuracy but also enhances the robustness of antimicrobial resistance (AMR) prediction, especially in cases where original datasets are limited. For data augmentation, EX2 utilizes an over-sampling approach known as SMOTE (Synthetic Minority Over-sampling Technique) [7] to balance the class distribution and enhance model generalization.

Figure 4 presents the performance of the CNN model across various genome-antibiotic combinations using 200 original samples, followed by different levels of data augmentation with 200, 400, and 600 additional synthetic instances. In the x-axis labels, “Original” refers to the original 200 samples, while “Aug200” represents the combination of the original 200 and 200 augmented samples. An exception is the *E. faecium* – ampicillin (AMP) pair, which utilizes only 100 original samples for training. This is because the susceptible class contains a total of 209 instances, and a portion of the data was reserved exclusively for testing to ensure a strict separation between training and testing datasets.



**Figure 4: CNN F1-score on Genome–Antibiotic Combinations with Augmented Data vs. VanOeffelen Baseline**

The experimental results reveal a consistent and compelling trend: data augmentation substantially enhances the performance of the CNN model across all genome–antibiotic combinations. Starting with a baseline of 200 original samples, each dataset is progressively augmented with 200, 400, and 600 synthetic instances. Across this range, all key evaluation metrics—Precision, Recall, F1-score, and Accuracy—demonstrate notable improvements with increasing levels of augmentation. In most cases, the stepwise addition of augmented data leads to gradual yet significant gains. For example, the *E. coli* – trimethoprim/sulfamethoxazole (*SXT*) pair shows an increase in accuracy from 0.7383 (Original) to 0.9668 (Augmented 600). Similarly, the *E. faecium* – ampicillin (*AMP*) pair achieves a perfect F1-score (1.0000) and near-perfect accuracy (0.9982) with 600 augmented samples. However, experimental results indicate that after a certain level of data augmentation, only marginal improvements are observed. This suggests that the effectiveness of augmentation tends to plateau beyond a certain threshold of synthetic data. Compared to VanOeffelen *et al.*'s XGBoost approach [26], EX2's CNN model—augmented with synthetic training samples—achieves consistently higher accuracy and reliability in AMR prediction. In addition, these results outperform previously reported outcomes

on similar genome–antibiotic pairs [4]. These findings demonstrate the practical benefits of integrating data augmentation with deep learning in genomics-based machine learning pipelines, as models trained with augmented data consistently outperform those trained solely on original samples—particularly in scenarios involving class imbalance or limited data availability.

### 4.3 Experiment 3 (EX3): Prediction on Unseen Bacteria

Building upon EX2, we design a multi-class antibiotic prediction model as a proof-of-concept. Based on the results shown in Figure 4, we apply selection criteria that prioritize combinations with high F1-scores and ensure no overlap between genomes and antibiotics. The optimal pairs identified for EX3 are: (1) *E. faecium* & VAN, (2) *E. coli* & AMP, (3) *A. baumannii* & AMK, and (4) *S. enterica* & CHL. Following the feature generation algorithm described in Section 3, we regenerate the features using only the susceptible files. Features are generated and the model is trained using 200 data samples from each of the four categories. A total of 1,000 features are selected from the top 10,000 most frequent features based on feature importance methods, as outlined in Step 6. For evaluation,

40 unseen samples per category, excluded from feature generation, are used to represent newly identified bacteria.

**Table 3: Comparison of Performance Metrics Between RF and CNN.**

| Metrics   | RF<br>(Original 200) | CNN<br>(Original 200) |             |             |             |
|-----------|----------------------|-----------------------|-------------|-------------|-------------|
|           |                      | +Aug<br>0             | +Aug<br>200 | +Aug<br>400 | +Aug<br>600 |
| Accuracy  | 1.0000               | 0.9938                | 0.9938      | 0.9938      | 1.0000      |
| Precision | 1.0000               | 0.9939                | 0.9939      | 0.9938      | 1.0000      |
| Recall    | 1.0000               | 0.9938                | 0.9938      | 0.9938      | 1.0000      |
| F1-score  | 1.0000               | 0.9937                | 0.9937      | 0.9937      | 1.0000      |

Table 3 presents a comparison of performance metrics between RF and CNN models. “+Aug” in the Table denotes the amount of augmented data added to the original dataset. For instance, “+Aug 600” indicates that 600 augmented samples were added to the original 200 training samples. The RF model, trained on the original 200 samples, achieved perfect classification performance, attaining 1.0000 across all metrics including Accuracy, Precision, Recall, and F1-score. In comparison, the CNN initially demonstrated slightly lower performance when trained without augmented data (+Aug 0), with metric values ranging from 0.9938 to 0.9939. In fact, only one misclassification occurred out of the 160 testing samples. Progressive increases in augmented data led to incremental performance gains for the CNN model. Notably, while adding 200 and 400 augmented samples maintained similar performance levels, it was only after adding 600 augmented samples that the CNN model achieved parity with the RF model, recording perfect scores. This trend underscores the importance of data augmentation in deep learning models such as CNNs, especially when operating in limited data environments, and contrasts with the RF model’s inherent stability and effectiveness with smaller datasets.

**Table 4: Structure of the Deep 1D CNN Used in Our Experiments**

| Layer Type     | Parameters                                  | Output Shape                 |
|----------------|---------------------------------------------|------------------------------|
| Input Layer    | Input shape: $(F, 1)$                       | $(F, 1)$                     |
| <i>Block 1</i> |                                             |                              |
| Conv1D         | 32 filters, kernel size 5, LeakyReLU(0.01)  | $(F, 32)$                    |
| BatchNorm1D    | –                                           | $(F, 32)$                    |
| MaxPooling1D   | pool size 2                                 | $(\lfloor F/2 \rfloor, 32)$  |
| <i>Block 2</i> |                                             |                              |
| Conv1D         | 64 filters, kernel size 5, LeakyReLU(0.01)  | $(\lfloor F/2 \rfloor, 64)$  |
| BatchNorm1D    | –                                           | $(\lfloor F/2 \rfloor, 64)$  |
| MaxPooling1D   | pool size 2                                 | $(\lfloor F/4 \rfloor, 64)$  |
| Dropout        | rate 0.2                                    | $(\lfloor F/4 \rfloor, 64)$  |
| <i>Block 3</i> |                                             |                              |
| Conv1D         | 128 filters, kernel size 5, LeakyReLU(0.01) | $(\lfloor F/4 \rfloor, 128)$ |
| BatchNorm1D    | –                                           | $(\lfloor F/4 \rfloor, 128)$ |
| Flatten        | –                                           | $(-)$                        |
| Dense          | 64 units, LeakyReLU(0.01)                   | 64                           |
| BatchNorm1D    | –                                           | 64                           |
| Dropout        | rate 0.4                                    | 64                           |
| Dense (Output) | 4 units, softmax                            | 4                            |

Shapes assume Conv1D uses *same* padding and stride = 1; MaxPooling1D uses pool size 2, stride 2. Sequence length  $F$  uses  $\lfloor \cdot \rfloor$  to denote integer division.

As detailed in Table 4, the deep 1D CNN, comprising three convolutional blocks followed by fully connected layers, achieved perfect performance (1.0000 across all metrics) when trained on the 200-sample dataset. The only trade-off compared with the standard CNN was a moderate increase in training time (451.70 seconds vs. 421.25 seconds), illustrating the balance between enhanced complexity and computational efficiency.

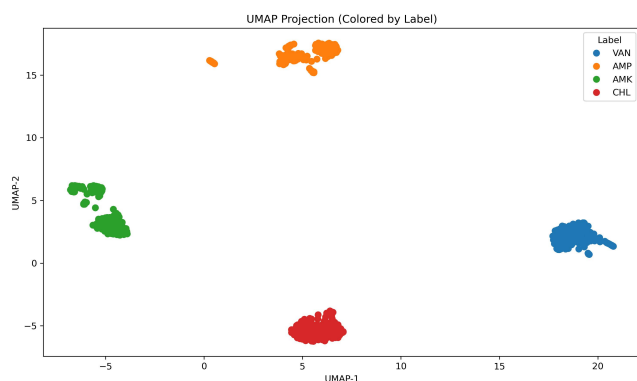
To assess the model’s practicality in real-world scenarios, we further measure its inference speed. In a clinical setting, the process of collecting bacterial samples and converting them into .fna files generally takes between 1.5 to 4 days. Even in high-throughput laboratories with optimized pipelines, the process still requires 24 to 36 hours. In contrast, once a moderate amount of original genomic data is obtained, our models can immediately generate synthetic data through augmentation techniques such as SMOTE, without incurring further delays related to sequencing or processing. This capability allows the model to be trained quickly, improving robustness in low-data settings without a significant computational burden during inference. As a result, our approach is highly practical for integration into clinical diagnostic pipelines, enhancing prediction performance without disrupting the overall timeline. Assuming that the bacterial samples obtained from patients have already been converted to .fna format in the laboratory, we evaluate the time required for downstream processing, including pre-processing and antibiotic prediction. Using a test set of 160 samples, the model achieves an average prediction time of 3.05 seconds. The time measurement experiments are performed on a system running macOS Sequoia (Version 15.4.1), equipped with an Apple M3 Pro processor featuring 11 cores (5 performance and 6 efficiency cores) and 18 GB of memory.

Figure 5 presents a two-dimensional UMAP (Uniform Manifold Approximation and Projection) projection of the dataset, with points colored by their annotated antibiotic resistance class. The visualization reveals four distinct and well-separated clusters corresponding to VAN, AMP, AMK, and CHL. The clear spatial separation indicates that the high-dimensional features effectively capture discriminative signals, resulting in strong inter-class differences and compact intra-class similarity. This outcome underscores that the feature engineering process was successful, as the representations extracted from the genomic data preserve biologically meaningful patterns, thereby enabling UMAP to project them into a low-dimensional space without loss of class separability. Consequently, the observed clustering validates the quality of the constructed features and provides strong evidence for their suitability in classification and prediction tasks.

## 5 Discussion

The experimental results presented in this study provide several key insights into the strengths and limitations of different machine learning strategies for antimicrobial resistance (AMR) phenotype prediction. By systematically evaluating tree-based ensemble models, convolutional neural networks (CNNs), and the integration of data augmentation techniques, we identify practical trade-offs that can inform the design of predictive pipelines in both research and clinical settings.





**Figure 5: UMAP Visualization of Selected Features Demonstrating Class Separability**

**Tree-based models in low-data environments.** Experiment 1 (EX1) demonstrated that tree-based models, particularly Random Forest (RF), Extra Trees (ET), and XGBoost (XGB), are highly effective in scenarios with limited data availability [16, 26]. These models consistently achieved strong performance even when restricted to a subset of 200 resistant and 200 susceptible samples. Notably, their success was amplified when paired with hyperparameter optimization methods, although no single tuning strategy (Bayesian, Grid, or Randomized) consistently outperformed the others. This observation emphasizes the need for model-specific and dataset-specific optimization rather than a one-size-fits-all tuning approach. Importantly, the strong results of tree-based models indicate that laboratories with modest sequencing resources can still leverage predictive modeling without the computational overhead of deep learning, which is a critical consideration for real-world deployment in resource-constrained environments.

**Data augmentation as a catalyst for deep learning.** Experiment 2 (EX2) highlighted the transformative role of data augmentation in enabling CNNs to outperform traditional ensemble models [2, 7, 8, 20]. As shown in Figure 4, the progressive addition of synthetic data (via SMOTE) resulted in significant improvements across accuracy, precision, recall, and F1-score. For instance, the *E. coli*–SXT combination improved from 0.7383 accuracy on the original dataset to 0.9668 after 600 augmented samples were added. These results underscore the utility of augmentation not merely as a means of balancing classes but also as a mechanism for enhancing the representational diversity of the training set. However, the plateau effect observed at higher augmentation levels suggests diminishing returns, implying that augmentation should be applied adaptively. Future research could explore generative models such as variational autoencoders or diffusion models to simulate biologically realistic mutations, potentially surpassing the capabilities of purely statistical oversampling techniques like SMOTE.

**Generalization to unseen bacteria.** Experiment 3 (EX3) provided a proof-of-concept for predicting resistance in unseen bacterial strains through multiclass classification. The ability of both RF and CNN models to achieve near-perfect scores across multiple antibiotic classes suggests that the proposed feature engineering pipeline preserves discriminative signals relevant for generalization.

The UMAP clustering in Figure 5 further confirmed that selected features form biologically meaningful and well-separated clusters, supporting the robustness of the feature selection process. However, it is important to acknowledge that the multiclass dataset was relatively small and limited to four genome–antibiotic pairs. While the models achieved perfect accuracy in this controlled setting, scaling to a broader range of pathogens and antibiotics will likely reveal greater variability. Incorporating domain-specific biological knowledge, such as resistance gene databases or mobile genetic element annotations, may be necessary to ensure reliable generalization at scale.

**Clinical applicability and practical considerations.** From a translational perspective, the inference time of our models (3.05 seconds for 160 samples) is negligible compared to the laboratory processes of bacterial culture and sequencing, which typically span multiple days. This efficiency positions machine learning models as highly compatible with diagnostic pipelines, where rapid turn-around can influence antibiotic stewardship decisions. Nevertheless, before clinical integration, several challenges must be addressed. These include external validation on independent cohorts, regulatory approval for clinical decision support, and strategies to ensure patient data privacy and security when handling genomic information. Additionally, while SMOTE was sufficient for this study, the adoption of biologically constrained augmentation could improve clinical trust by generating synthetic data that more closely mimics real resistance mechanisms.

**Limitations and future directions.** While this study makes important contributions, several limitations should be noted. First, the analysis focused on eight genome–antibiotic pairs; expanding to a broader spectrum of bacterial species and resistance phenotypes will be necessary to confirm the generalizability of the approach. Second, we limited augmentation methods to SMOTE; alternative strategies such as GAN-based sequence generation or hybrid augmentation could be explored. Third, although CNNs demonstrated strong performance, additional deep learning architectures such as transformers or graph neural networks (GNNs) may capture longer-range genomic dependencies more effectively than k-mer features alone. Finally, this study primarily evaluated predictive performance; integrating model interpretability frameworks (e.g., SHAP or attention-based visualization) would provide deeper biological insights and increase trust in clinical settings.

**Summary of key insights.** Taken together, the results suggest that a hybrid strategy offers the most promise for AMR prediction: (1) tree-based models for rapid, low-data scenarios, and (2) deep learning models augmented with synthetic data for high-accuracy, large-scale applications. By explicitly balancing the trade-offs between data availability, computational cost, and predictive performance, the proposed framework provides a pathway toward practical, robust, and clinically meaningful AMR prediction. These findings also highlight the broader lesson that methodological flexibility—rather than reliance on a single model or technique—is critical for tackling the heterogeneous and data-scarce challenges characteristic of AMR research.

## 6 Conclusion

Accurately predicting antimicrobial resistance (AMR) remains a major challenge due to the limited availability of labeled genomic data and the high computational cost of feature extraction and preprocessing. Although existing machine learning models have shown promise, their performance is often constrained by data imbalance, feature sparsity, and limited generalizability across diverse genome–antibiotic combinations. This study demonstrated that combining  $k$ -mer-based feature extraction, data augmentation, and deep learning substantially improves AMR prediction accuracy, particularly in data-scarce and imbalanced settings. In EX1 and EX2, the CNN model, trained on synthetic datasets composed of both original and augmented samples, consistently outperformed or matched previous benchmarks. Tree-based models achieved efficient and robust performance, especially on smaller datasets. Furthermore, EX3, a multiclass setting with novel bacteria, confirmed the model's generalizability by accurately distinguishing antibiotic classes, showing feature and model robustness on unseen data.

The results emphasize the effectiveness of integrating data augmentation with both deep learning and traditional machine learning methods to overcome key limitations in AMR prediction. One limitation of this study is that the models were evaluated on a selected set of genome–antibiotic combinations, which may not fully capture the diversity and complexity of resistance mechanisms across all bacterial species. Additionally, another limitation is that we did not compare the use of other data augmentation algorithms beyond SMOTE or explore alternative models beyond CNN. Future research should expand the coverage of genome–antibiotic combinations to include a broader spectrum of bacterial species and resistance phenotypes, improving the robustness and applicability of the models. Incorporating real-world clinical genomic datasets, exploring advanced data augmentation strategies and various deep learning models including multi-class classification may further enhance prediction accuracy.

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